



Lab 01

Course: Operating System

Class: AI Section A and B

Instructor: Bilal Ahmed

Learning Objectives:

- Installation of Virtual Box (Optional).
- Installation of Ubuntu.
- Basic commands in Linux.
- Write and execute python code in Linux terminal.

Introduction: An operating system (OS) is software that acts as an intermediary between computer hardware and the computer user. It provides a platform for other software to run and manage system resources, such as memory, processors, and devices. The primary functions of an operating system include process management, memory management, file system management, and user interface management.

Example: Windows, Linux, Android etc.

Reasons to work with Ubuntu: When it comes to experimenting with operating system concepts, Ubuntu (Linux) often has distinct advantages due to its open-source nature, robust command-line interface, and community-driven development.

Installation:

Download Latest Ubuntu.

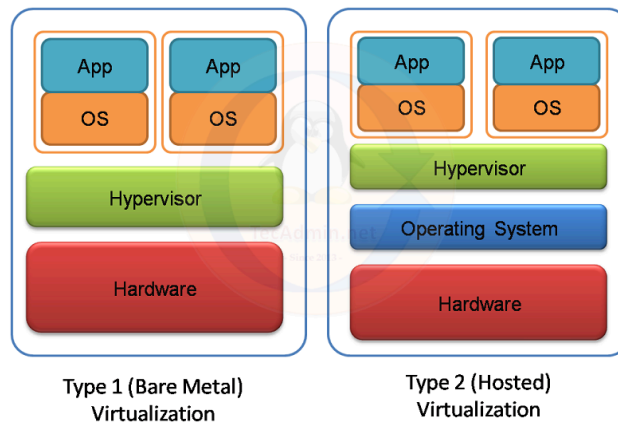
Link: <https://ubuntu.com/download/desktop>

Download the Latest Virtual Box.

Link: <https://www.virtualbox.org/wiki/Downloads>

Virtual Box: Oracle VM VirtualBox is a type-2 hypervisor for x86 virtualization developed by Oracle Corporation.

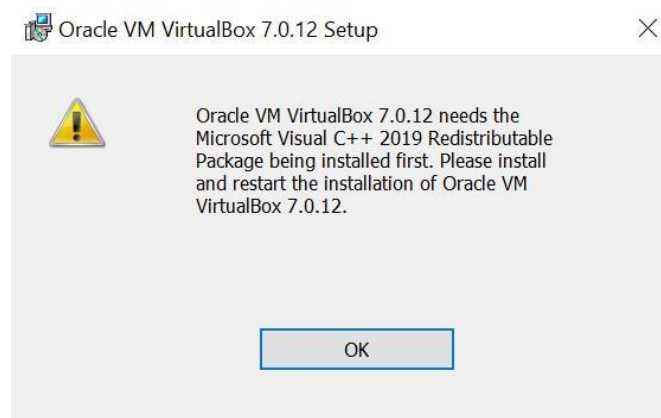
Hypervisor: A hypervisor is a type of computer software, firmware or hardware that creates and runs virtual machines. A computer on which a hypervisor runs one or more virtual machines is called a host machine, and each virtual machine is called a guest machine.



Virtual Machine: A VM is a virtualized instance of a computer that can perform almost all of the same functions as a computer, including running applications and operating systems. Virtual machines run on a physical machine and access computing resources from software called a hypervisor.

VMware vs Virtual Box: VirtualBox is open-source and, therefore, free to use. VMware is only free for personal or educational use, and it has limited functionalities. Virtual machines created by VMware are much faster than those created by VirtualBox.

Problem while installing virtual box: While Installing Virtual Box you may encounter the following error.



Solution: Install the Microsoft Visual C++ 2019 Redistributable Package for your Computer.



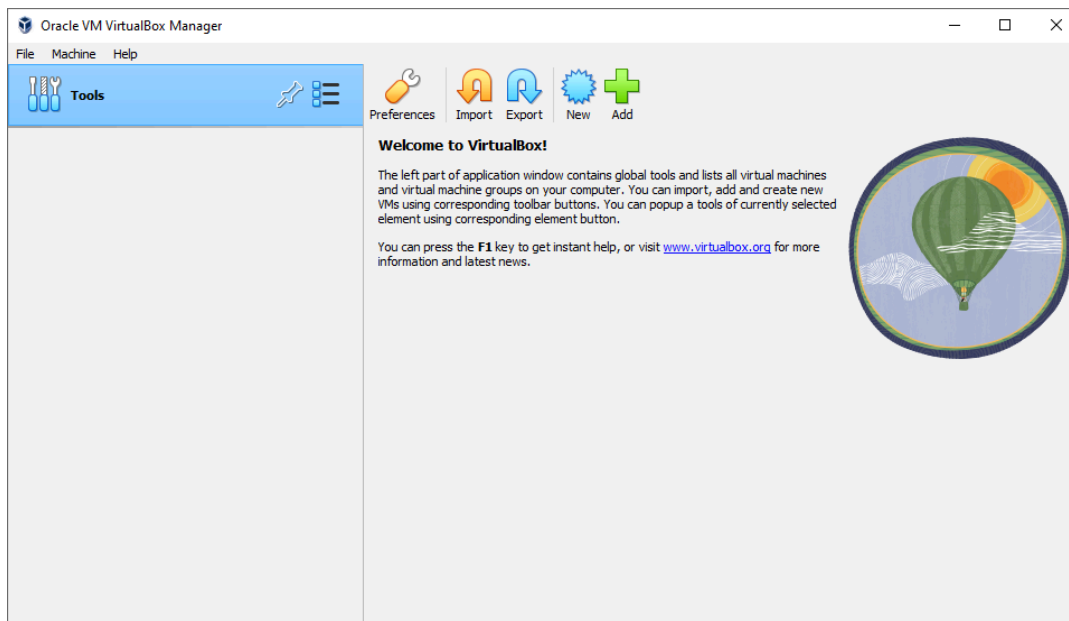
Architecture	Link	Notes
ARM64	https://aka.ms/vs/17/release/vc_redist.arm64.exe	Permalink for latest supported ARM64 version
X86	https://aka.ms/vs/17/release/vc_redist.x86.exe	Permalink for latest supported x86 version
X64	https://aka.ms/vs/17/release/vc_redist.x64.exe	Permalink for latest supported x64 version. The X64 Redistributable package contains both ARM64 and X64 binaries. This package makes it easy to install required Visual C++ ARM64 binaries when the X64 Redistributable is installed on an ARM64 device.

Link:

<https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170>

Direct Link for x64: https://aka.ms/vs/17/release/vc_redist.x64.exe

Virtual Box Interface: After installation of Virtual Box, you will see following.



Installing Ubuntu: Click Machine and click the new button to add machine (Ubuntu in this case). You will have the following output.



Adding path: Give the hostname in name field and just select the ISO Image Dropdown and give the path of ISO file of Ubuntu and everything else will be set accordingly as given below.

Setting credentials: Give your username and password as I have given Ubuntu here.



Create Virtual Machine

Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Username: ✓

Password: ✓

Repeat Password: ✓

Additional Options

Product Key:

Hostname: ✓

Domain Name:

☐ Install in Background

☐ Guest Additions

Guest Additions ISO:

Help Back Next Cancel

Allocating Virtual Memory and Virtual CPU: Allocate RAM and number of CPUs according to your need and specifications for your system. I have given 4GB memory and 4 CPUs as shown below.

Create Virtual Machine

Hardware

You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count. Enabling EFI is also possible.

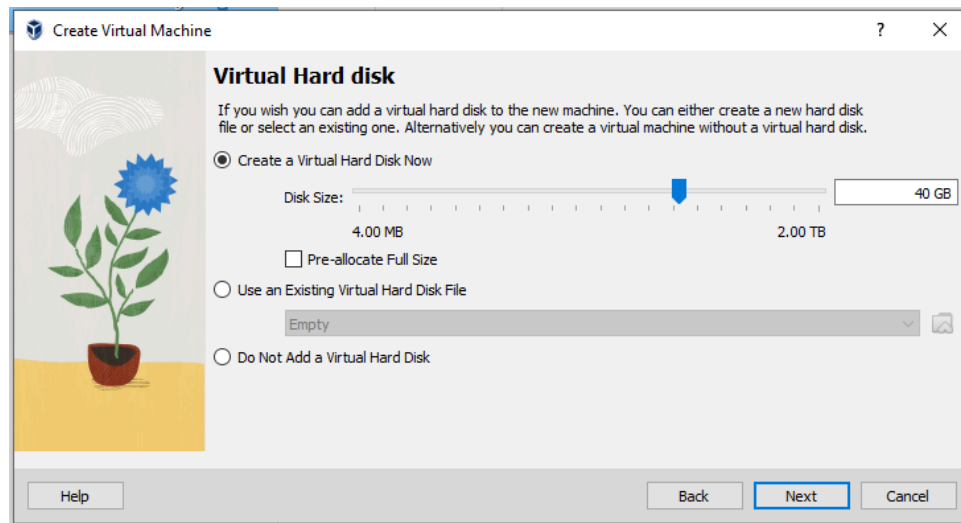
Base Memory: 4096 MB

Processors: 4

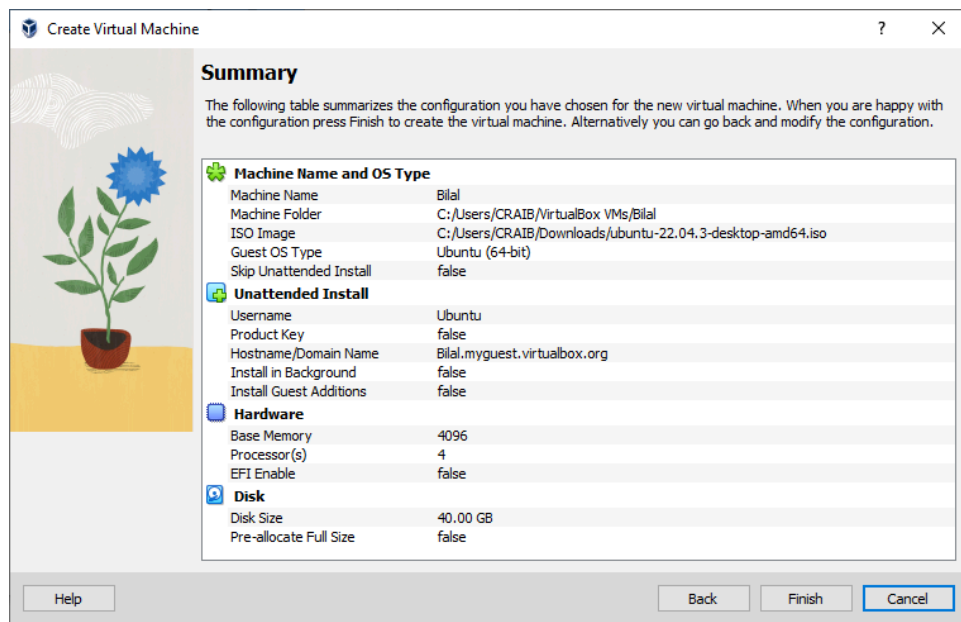
☐ Enable EFI (special OSes only)

Help Back Next Cancel

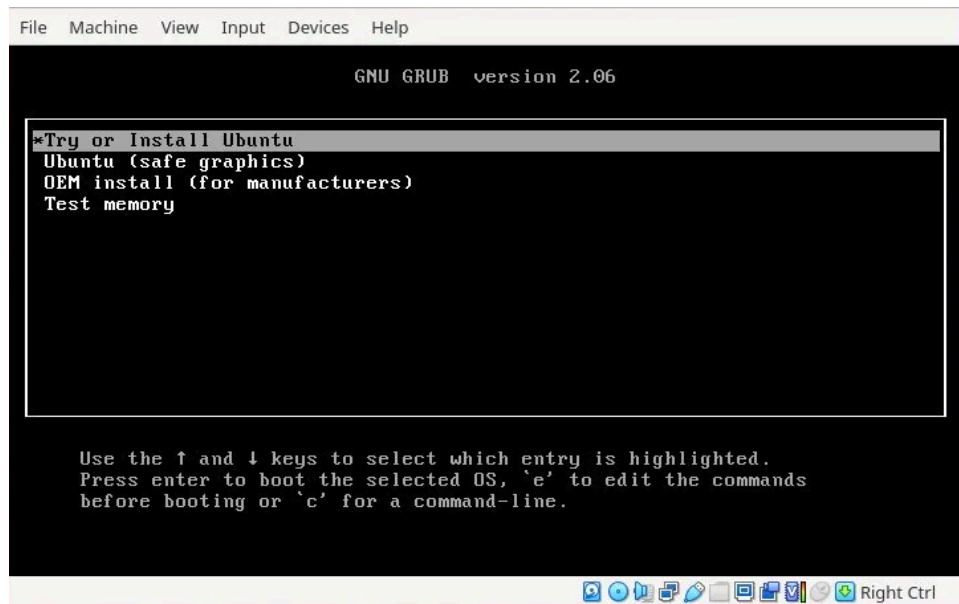
Allocating Virtual Hard disk: Allocate virtual hard disk as I have allocated 40GB below.



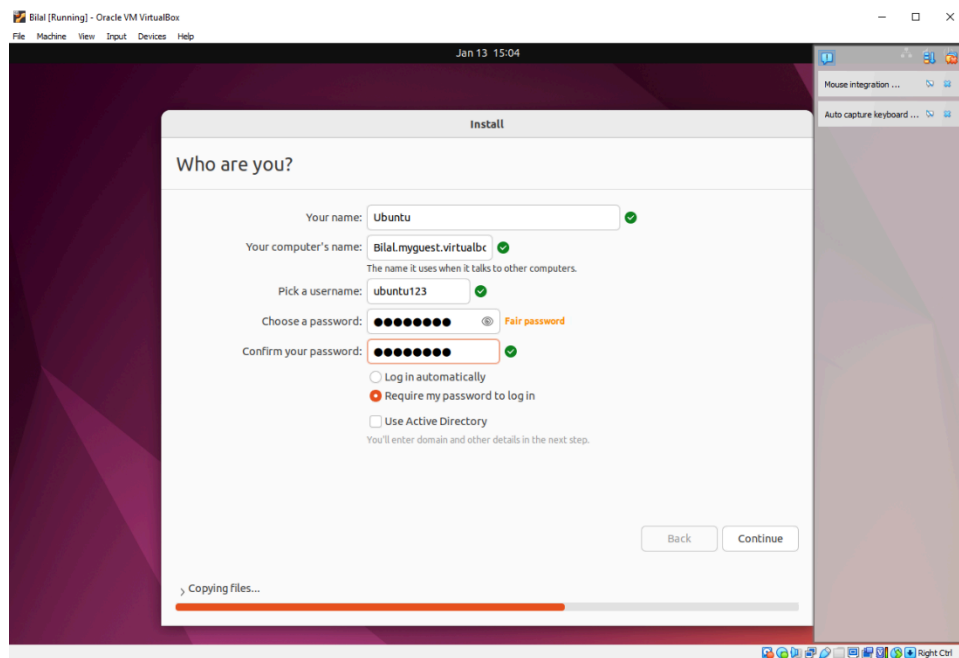
VM Summary: Once everything is done, it will print the summary of VM to be installed as below.



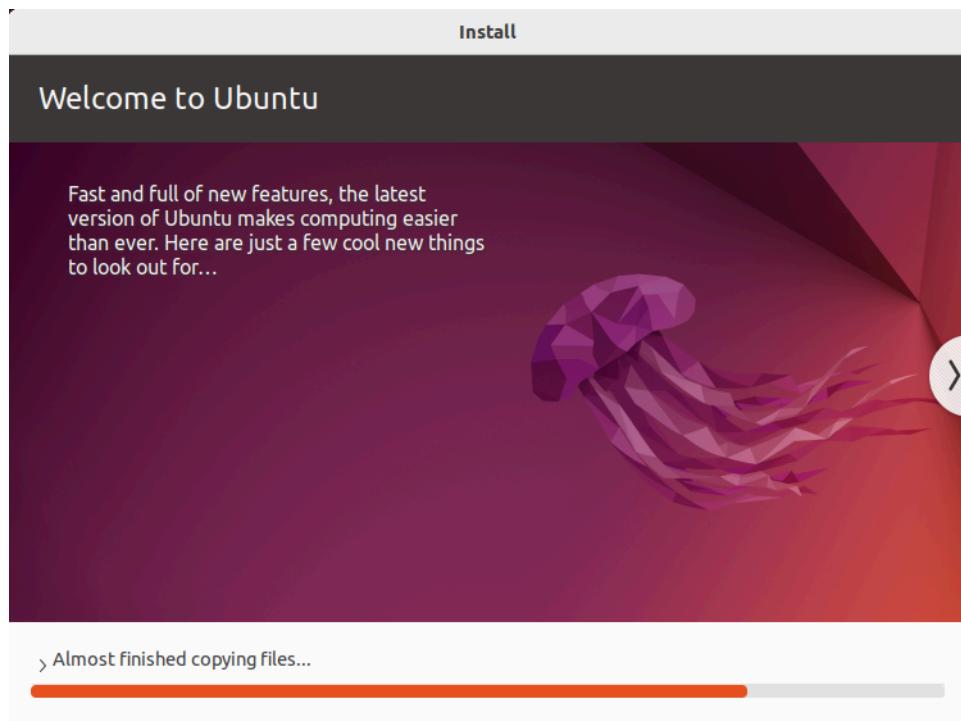
Install Ubuntu: Make Sure Virtualization is enabled in BIOS.



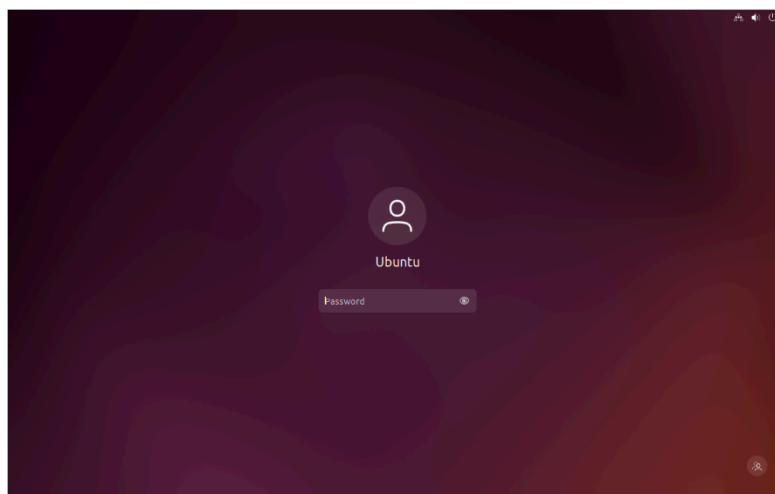
Set Credentials: Give Username and password as given below and click the continue.



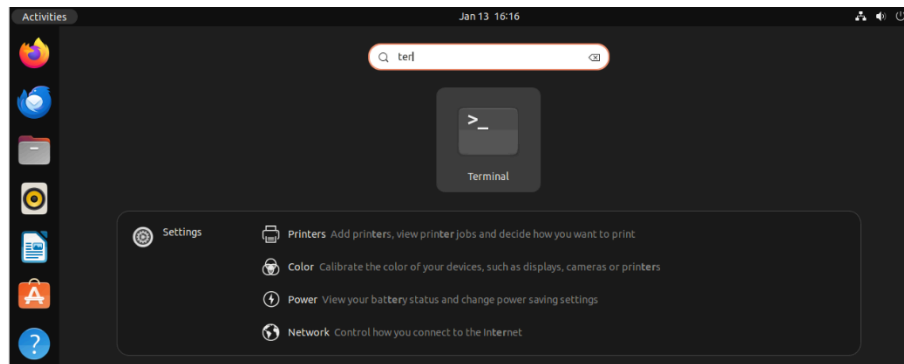
Wait for installation:



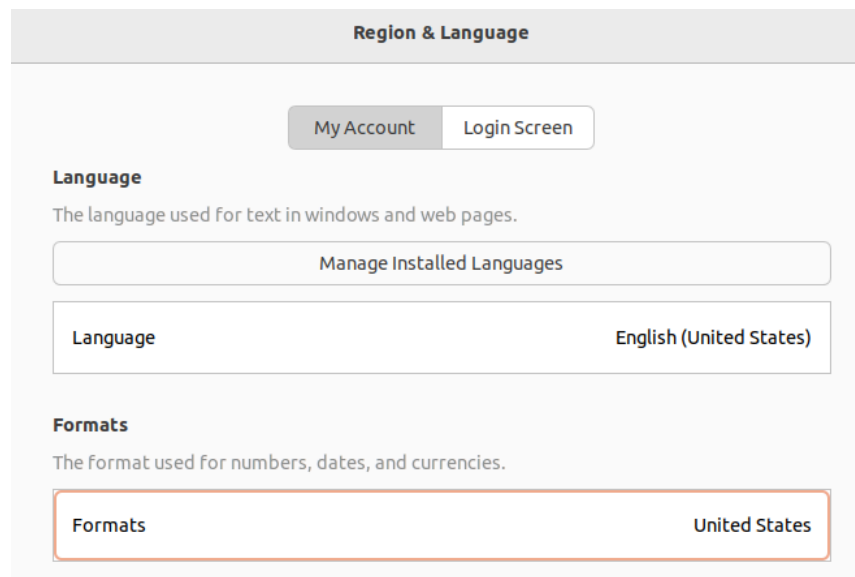
Installation Complete: If everything worked fine, you will be come up with the following window waiting for log in.



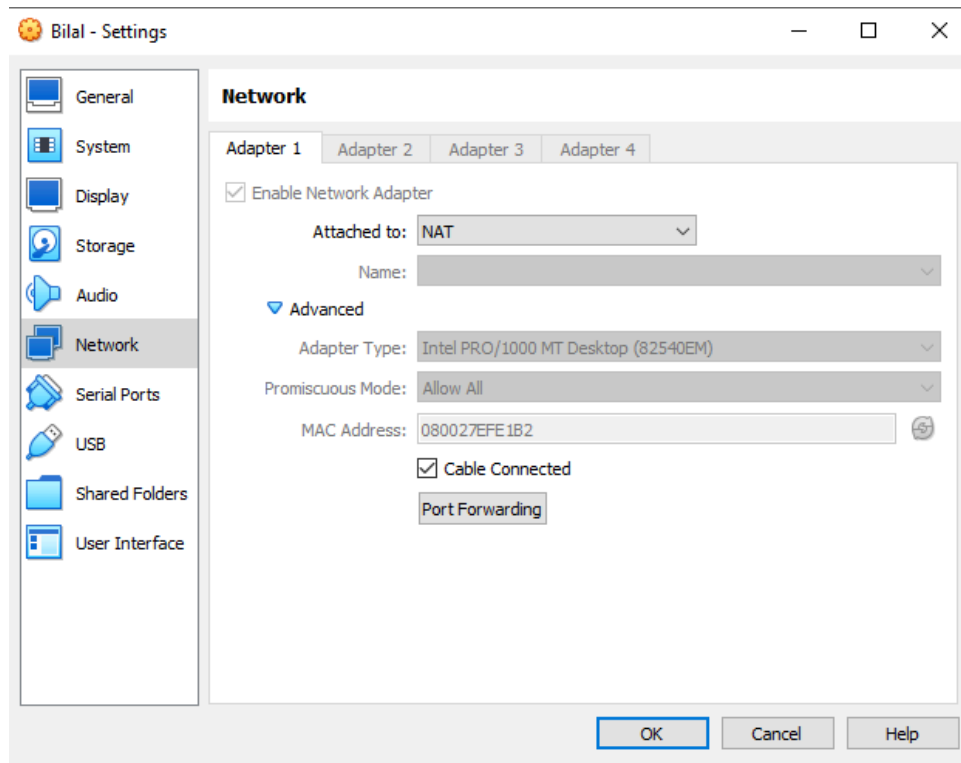
Testing the terminal: Open the terminal with *ctrl+shift+t* or search by pressing windows key as given below.



Terminal Issue: If terminal doesn't open go to **Settings>Region & Language>** in Ubuntu installed in Virtual Box and set all settings to United States as below. And reboot the machine.



Network Settings: The Internet should be working and if it doesn't go to the Devices option in the Virtual Box running machine, attach network adapter to NAT, go to advanced and Promiscuous Mode to allow all if it is not already as shown below.



Update the system: Update the system by using **sudo apt update** command in the terminal as below. Sudo works here as an administrator.

```
ubuntu123@Bilal: ~  
ubuntu123@Bilal:~$ sudo apt update  
[sudo] password for ubuntu123:  
Hit:1 http://us.archive.ubuntu.com/ubuntu jammy InRelease  
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
180 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu123@Bilal:~$
```

Basic Linux Commands.



1. **pwd - Print Working Directory:**

- *Description:* Displays the current working directory (the directory you are currently in).

```
jackal@jackal: ~  
jackal@jackal:~$ pwd  
/home/jackal  
jackal@jackal:~$
```

2. **ls - List:**

- *Description:* Lists files and directories in the current directory.

```
jackal@jackal:~/Desktop$ ls  
CODE_OFF_Gerber  eagle  Jackal_Pics  
CODE_OFF_Gerber.zip  'Ibadullah CV.pdf'  'My Library.rdf'  
jackal@jackal:~/Desktop$
```

3. **cd - Change Directory:**

- *Description:* Changes the current working directory. For example, cd Documents would move to the "Documents" directory.

```
jackal@jackal: ~/Desktop/Jackal_Pics  
jackal@jackal:~$ cd Desktop/  
jackal@jackal:~/Desktop$ cd Jackal_Pics/  
jackal@jackal:~/Desktop/Jackal_Pics$
```

4. **mkdir - Make Directory:**

- *Description:* Creates a new directory. For example, mkdir NewFolder creates a new directory named "NewFolder."

```
jackal@jackal: ~/Desktop  
jackal@jackal:~$ cd Desktop/  
jackal@jackal:~/Desktop$ mkdir Computer  
jackal@jackal:~/Desktop$ ls  
CODE_OFF_Gerber  Computer  'Ibadullah CV.pdf'  'My Library.rdf'  
CODE_OFF_Gerber.zip  eagle  Jackal_Pics  
jackal@jackal:~/Desktop$
```

5. **rmdir - Remove Directory:**

- *Description:* Removes an empty directory. Use with caution, as it deletes the directory without confirmation.

6. **touch - Create Empty File:**



- *Description:* Creates an empty file. For example, touch newfile.txt creates a new empty file named "newfile.txt."

Python Program: Creating a simple python program using nano editor command and printing hello world.

```
bilal@Bilal: ~  
bilal@Bilal:~$ nano simple.py  
bilal@Bilal:~$
```

Saving program: After writing press **ctrl + x**, press **y** and **enter** to save it.

```
GNU nano 4.8 simple.py Modified  
print("Hello World")  
  
File Name to Write: simple.py  
^G Get Help      M-D DOS Format  M-A Append      M-B Backup File  
^C Cancel        M-M Mac Format  M-P Prepend     ^T To Files
```

Running: Write **python3 simple.py** in terminal to execute the program as below.

```
bilal@Bilal:~$ nano simple.py  
bilal@Bilal:~$ python3 simple.py  
Hello World  
bilal@Bilal:~$
```

Exercise:

Task 1: Ubuntu lies in the Debian family of Linux. Compare and discuss the following distributions with examples according to their usage.

- Debian
- RedHat
- SUSE
- Arch Linux
- Gentoo



Task 2: We used Virtual Box that is a type 2 hypervisor. Compare and discuss the type 1 and type 2 hypervisor with real life and well-known examples of each.

Task 3: Linux Commands attach screenshots of each step.

- Create a folder with your name.
- Create one sample python and one text file.
- Add some content in both files and run the python program.

Task 4: Explore and run the equivalent commands for **pwd** and **ls** in windows command line (DOS) and attach screenshots.